
SIGHT: AI-Driven Immersive Safety Training for Industry 5.0 Manufacturing

Dr. Arthur Carvalho
arthur.carvalho@miamioh.edu



MIAMI UNIVERSITY

Manufacturing is a cornerstone of the U.S. economy

**\$2.9
Trillion**

Annual contribution to
U.S. GDP.

**12.7
Million**

Workers employed in
the sector.

54%

Share of R&D conducted
by American businesses.

The background of the slide is a dark, industrial scene featuring large metal gears, belts, and machinery components. The lighting is dramatic, with highlights on the metallic surfaces and deep shadows in the recessed areas, creating a sense of a busy, potentially hazardous work environment.

Yet, persistent safety challenges carry a severe human and financial cost

>320,000

Workplace injuries reported annually

\$7.5 Billion

Estimated annual direct costs (medical payments and lost wages)

11 Days

Average lost workdays per non-fatal injury

Manufacturing

- Contributing factor
 - Manufacturing training is challenging
 - Trainees must:
 - Review machine manuals
 - Watch tutorials and demos
 - Listen to instructors



SIGHT

- **Safety Immersion and Gamified Hazard Training (SIGHT)**
 - \$1.5 million grant from the Ohio Bureau of Workers' Compensation (BWC) through its Worker Safety Innovation Center (WSIC)

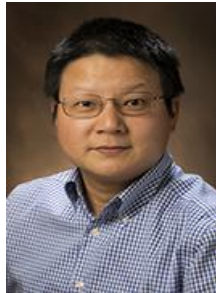
Dr. Fadel Megahed
Role: PI



Dr. Arthur Carvalho
Role: Co-PI



Dr. Jay Shan
Role: Co-PI



Dr. Mohammad Mayyas
Role: Co-PI



Dr. Reza Abrishambaf
Role: Sr. Personnel



Dr. Ibrahim Yousif
Role: Postdoc



Department of Information Systems and Analytics

meerkai

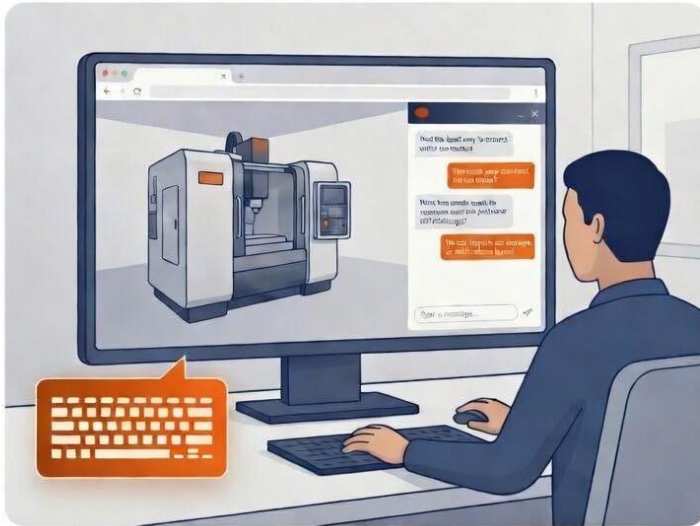
Department of Engineering Technology

maxbyte.
inspiring industry x.0

SIGHT

➤ Two manufacturing training solutions:

Simulation: VR + AI

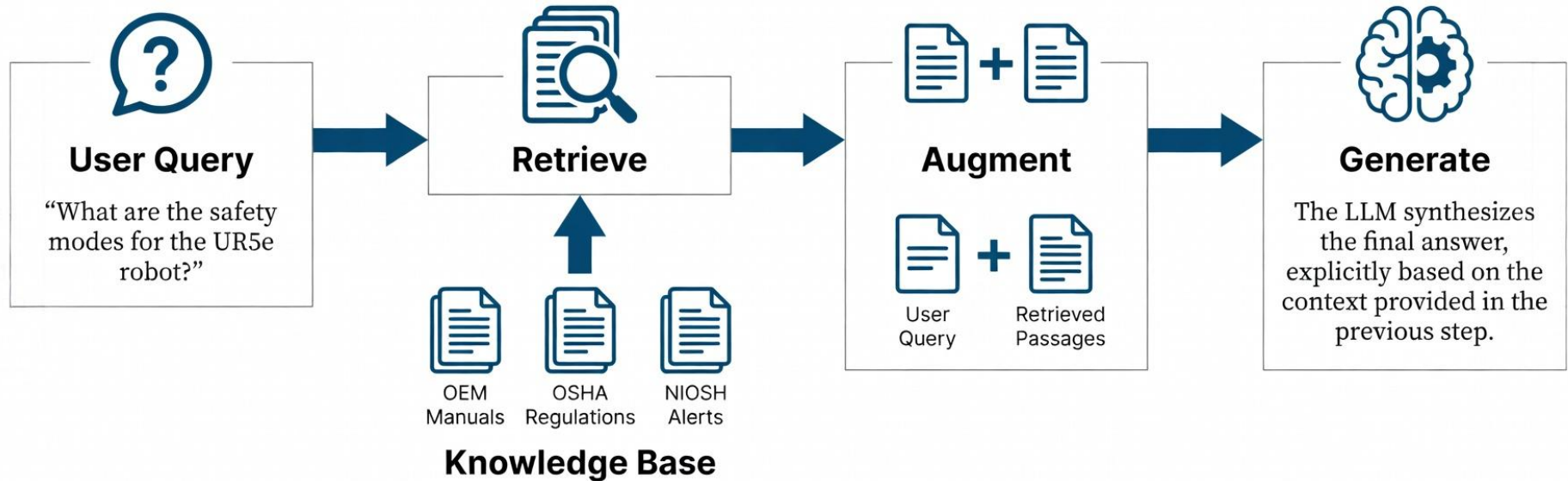


On the Floor: AR + AI



AI Component

➤ Retrieval Augmented Generation (RAG) technique



Experiments

➤ RAG pipelines

Factor	Levels
Retrieval Method	OpenAI Keyword OpenAI Semantic BM25 Graph Eager Graph MMR Vanilla Similarity
Large Language Model	gpt-5-mini-2025-08-07 gpt-5-nano-2025-08-07
Retrieval Depth	Top 3 Chunks (k=3) Top 7 Chunks (k=7)

Total Configurations Tested:
 $6 \times 2 \times 2 = 24$ Pipelines

Experiments

- Comprehensive benchmark of Q&A pairs
 - Derived from official operational and safety manuals



Bridgeport Manual Mill
Legacy Manual System
42 Q&A Pairs



Haas TL-1 CNC Lathe
Industry 3.0 Automated System
51 Q&A Pairs



Universal Robots UR5e Cobot
Industry 5.0 Collaborative System
60 Q&A Pairs

- Our results focus on the UR5e Cobot

Experiments

➤ Three metrics



Accuracy

Training artifacts must deliver information that is technically correct and operationally safe. Inaccurate guidance can lead to fatal accidents.



Latency

Artifacts must provide timely and responsive feedback. Delayed information is irrelevant in a dynamic environment.

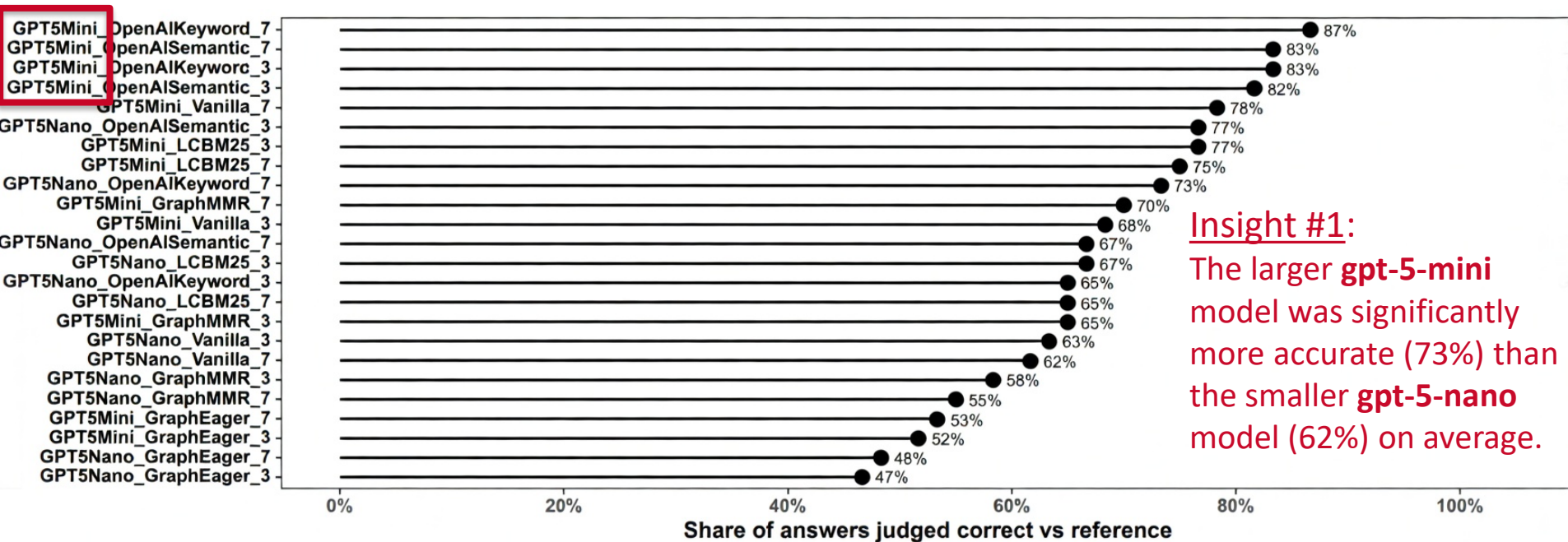


Cost

Artifacts must be affordable to ensure broad adoption, especially for the 98% of U.S. manufacturers that are small and medium-sized businesses.

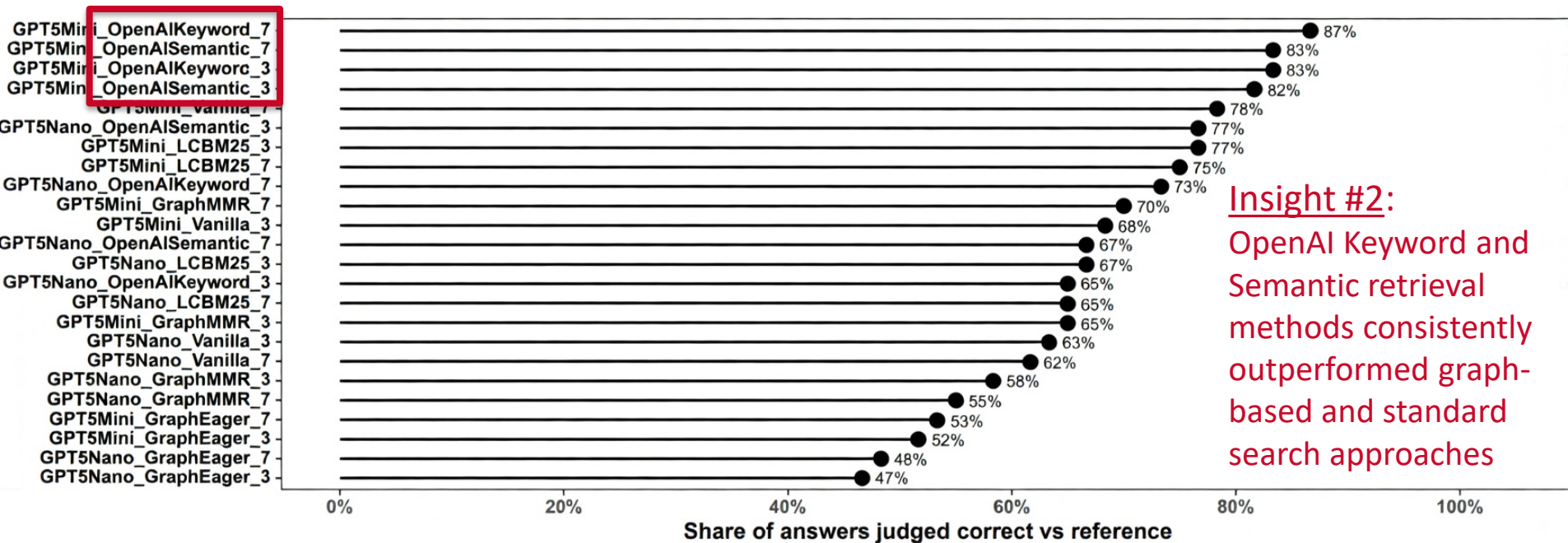
Results

➤ Accuracy: Model size and retrieval strategies are critical



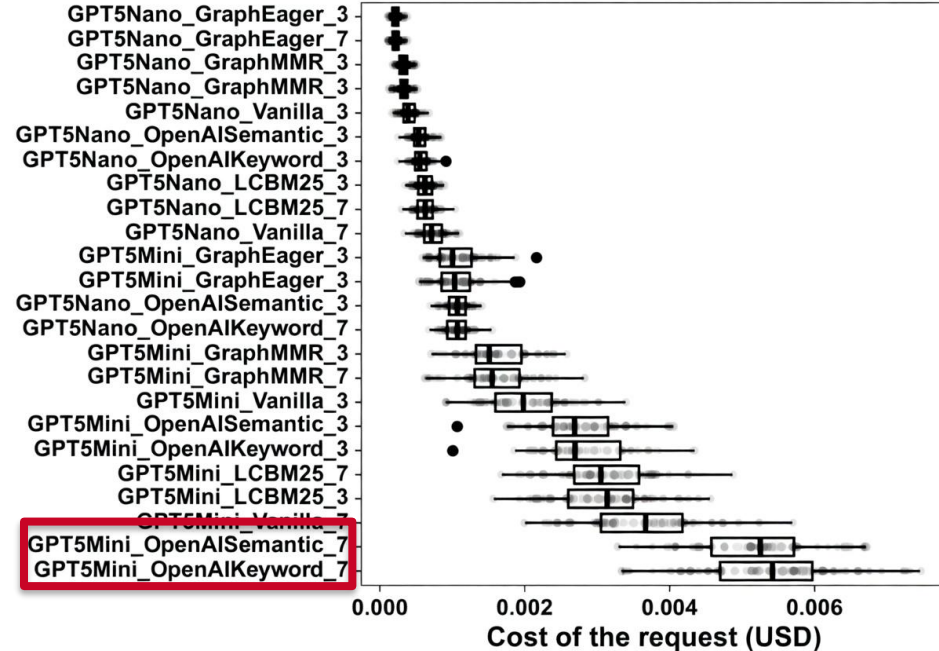
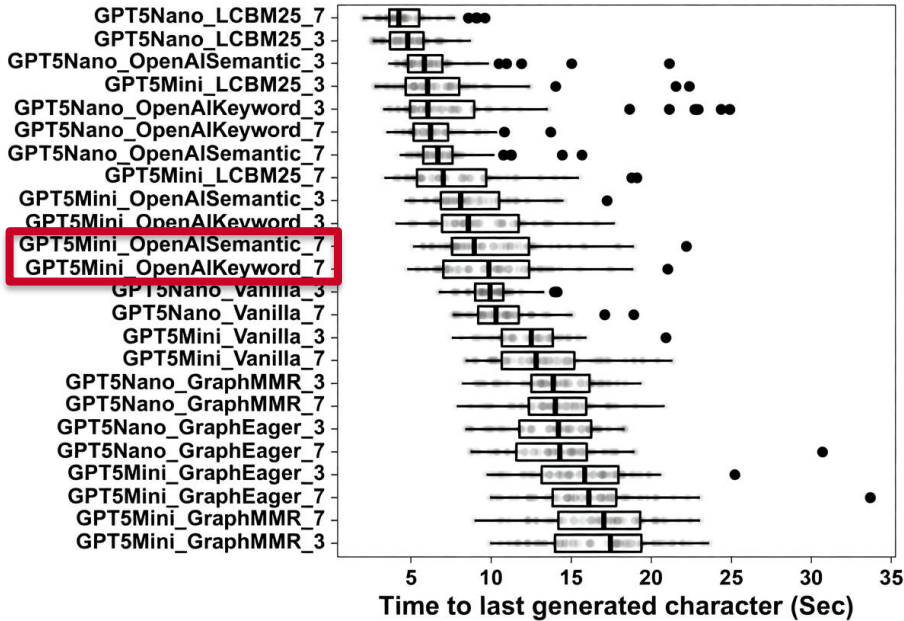
Results

➤ Accuracy: Model size and retrieval strategies are critical



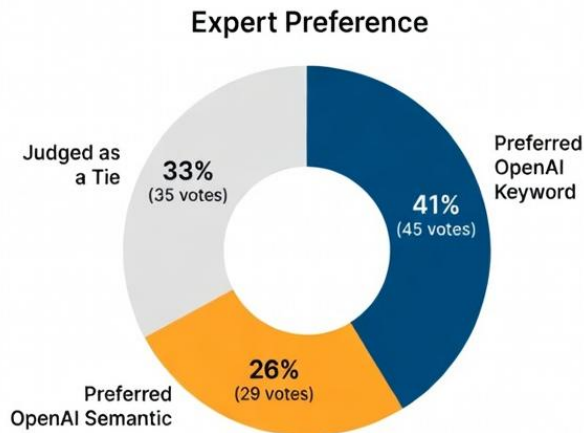
Results

➤ Latency and Cost: Trade-offs



Results

- No single pipeline is the best on all three metrics
 - Our approach: Maximize for accuracy
- **Final test:** Blind evaluation by experts
 - Top two pipelines based on accuracy



What safety modes does the robot arm have?

Generated Answer A

Generated Answer B

☒ Prefer A ☐ Prefer B ☐ Tie/Equal Quality

Expert evaluations
corroborate the
automated metrics

Results

➤ Final pipeline: Maximizing for accuracy

The Winning Build

Model: gpt-5-mini-2025-08-07

Retrieval Strategy: OpenAI Keyword

Retrieval Depth: top-k = 7

Accuracy

86.66%

Latency

**10.04 sec
(avg.)**

Cost

**\$0.005 / query
(avg.)**

Demonstration

Learning Objectives

- ✓ Read machine safety labels ⓘ
- ✓ Understand what are the danger zones ⓘ
- ✓ Power on the machine ⓘ
- ✓ Cycle the doors ⓘ
- ☐ Auto zero the machine ⓘ



POWER ON



POWER OFF



EMERGENCY STOP

ENTER

CANCEL

PROGRAM POSITION OFFSET CURRENT COMMANDS EDIT

ALARMS DIAGNOSTIC SETTINGS SETTINGS MEMORY

HOME



PAGE UP



CURSOR



END

Switch to the 3rd keyboard section of the panel

MOI

HANDLE JOG

ZERO RETURN



Miami University Bot
AI assistant



What is the purpose of the CNC machine?



- The CNC machine (Haas CNC lathe) is intended for cutting and shaping metals and other hard materials.

Type here...

